

Work № 3. STATE FEEDBACK ADAPTIVE CONTROL

Problem statement. Consider the plant presented in the “state-space” form¹

$$\dot{x} = Ax + bu, \quad x(0) \quad (3.1)$$

$$y = c^T x, \quad (3.2)$$

where $x \in R^n$ is the state vector, $u \in R$ is the control variable, $y \in R$ is the regulated variable, $A \in R^{n \times n}$ is an unknown constant matrix, $b, c \in R^n$ are the known constant vectors.

The problem is to design a control law that will compensate the influence of the parametric uncertainty on the closed-loop system stability and provide the limiting equality

$$\lim_{t \rightarrow \infty} \|x_m(t) - x(t)\| = \lim_{t \rightarrow \infty} \|e(t)\| = 0, \quad (3.3)$$

where $e = x_m - x$ is the vector of control errors, $x_m \in R^n$ is the state vector of reference model

$$\dot{x}_m = A_M x_m + b_M g, \quad (3.4)$$

$$y_m = c_M^T x_m \quad (3.5)$$

with a piece-wise continuous but bounded reference signal $g(t)$, a Hurwitz constant matrix $A_M \in R^{n \times n}$, and known constant vectors $b_M, c_M \in R^n$.

Note that in general case, the class of plants is restricted by the following assumption.

Assumption 3.1 (*Matching conditions*). There is an n - dimensional vector θ and a scalar κ such that the matrices A , b , A_m and b_m are related via equalities

$$A_M = A + b\theta^T, \quad b = \kappa b_M \quad (3.6)$$

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This assumption enables us to apply *certainty equivalence principle* and design an error model used for derivation of an adaptation algorithm. In this work, we assume that the triple (A, b, c) is given by canonical form

¹ The output equations of the plant model as well as the reference model below are given for the sake of completeness of the background presentation, however are not used in the state-feedback control derivations.

$$A = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-1} \end{bmatrix}, b = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ b_0 \end{bmatrix}, c = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

with unknown parameters $a_i, i=0,1,\dots,n-1$, and known constant b_0 , while the triple (A_m, b_m, c_m) is given by the same canonical form

$$A_M = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_{m0} & -a_{m1} & -a_{m2} & \cdots & -a_{mn-1} \end{bmatrix}, b_M = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ a_{m0} \end{bmatrix}, c_M = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

with the parameters a_{mi} ($i=0,1,\dots,n-1$) calculated using the pole-placement approach [1, 2] to provide the desired transient performance for the closed-loop system after adaptation process. In this case,

$$\theta_1 = \frac{-a_{m0} + a_0}{b_0}, \theta_2 = \frac{-a_{m1} + a_1}{b_0}, \dots, \theta_n = \frac{-a_{mn-1} + a_{n-1}}{b_0}, \kappa = \frac{b_0}{a_{m0}}. \quad (3.7)$$

Problem solution. Assuming the plant parameters known we design a control providing condition (3.3) with desired transient performance determined by transient time t_T and the overshooting $\bar{\sigma}$.

To this end, we define the control error as $e = x_m - x$ and calculate its time derivative in view of (3.1), (3.4) and (3.6):

$$\begin{aligned} \dot{e} &= \dot{x}_m - \dot{x} = A_M x_m + b_M g - Ax - bu = A_M x_m + b_M g - (A_M - b\theta^T)x - bu \\ &= A_M e + b \left(\frac{1}{\kappa} g + \theta^T x - u \right) \end{aligned}$$

or

$$\dot{e} = A_M e + b \left(\theta^T x - u + \frac{a_{m0}}{b_0} g \right), \quad (3.8)$$

which can be used for design of nonadaptive control

$$u = \theta^T x + \frac{a_{m0}}{b_0} g. \quad (3.9)$$

Replacing (3.9) in (3.8), we obtain the following control error model ensuring exponential convergence of the error e to zero:

$$\dot{e} = A_M e.$$

Since initially the parameters of the matrix A are unknown, then the control law (3.9) is not implementable. Therefore, let replace the unknown parameters θ by the adjustable parameters $\hat{\theta}(t)$ in (3.9) and obtain the adjustable control law

$$u = \hat{\theta}^T x + \frac{a_{m0}}{b_0} g. \quad (3.10)$$

Substitute the last equality into (3.8) and get the control error model

$$\dot{e} = A_M e + b \tilde{\theta}^T x, \quad (3.11)$$

where $\tilde{\theta} = \theta - \hat{\theta}$ is the vector of parametric errors.

Generalizing solution presented in the work №1 for multivariable case we select the Lyapunov function candidate

$$V = \frac{1}{2} e^T P e + \frac{1}{2\gamma} \tilde{\theta}^T \tilde{\theta},$$

where $P = P^T \succ 0$ is the solution of the Lyapunov equation

$$A_M^T P + P A_M = -Q \quad (3.12)$$

with preselected positively defined constant matrix Q . Calculating time derivative of V in view of (3.11) we obtain:

$$\dot{V} = -\frac{1}{2} e^T Q e + \tilde{\theta}^T x b^T P e + \frac{1}{\gamma} \tilde{\theta}^T \dot{\tilde{\theta}}.$$

Now, if we select the following structure for the adaptation algorithm

$$\dot{\hat{\theta}} = \gamma x b^T P e, \quad \hat{\theta}(0) = 0 \quad (3.13)$$

then the Lyapunov function derivative will be nonpositive, i.e.,

$$\dot{V} = -\frac{1}{2} e^T Q e \leq 0.$$

As a result, the choice (3.13) ensures the objective (3.3).

Note that the coefficient $\gamma > 0$ in (3.13) is called *adaptation gain* responsible for the rate of parametric convergence (3.10).

Thus the algorithm of adaptive control consists of adjustable control law (3.10) and adaptation algorithm (3.13), in which the matrix P is calculated using (3.12).

Under Assumption 3.1, for any initial conditions in the plant and the adaptation algorithm $x(0)$, $\hat{\theta}(0)$ and any bounded g adaptive control (3.10), (3.13) ensures [4-6]:

P.3.1. the boundedness of all the signals in the closed-loop system;

P.3.2. the convergence $e(t) \rightarrow 0$ as $t \rightarrow \infty$;

P.3.3. exponential convergence of $\tilde{\theta}(t)$ to zero if x satisfies the *persistent excitation condition* ($x \in PE$)

$$\int_t^{t+T} x(\tau)x^T(\tau)d\tau > \alpha I \quad (3.14)$$

for some constant values $\alpha > 0$, $T > 0$.

Condition (3.14) is equivalent to the existence of at least $n/2$ harmonics (spectral lines) in the signal $g(t)$;

P.3.4. if $\tilde{\theta}(t) \rightarrow 0$ as $t \rightarrow \infty$, then there exists an optimal coefficient γ , for which the rate of the convergence of $\tilde{\theta}(t) \rightarrow 0$ is maximum.

The order of the work

1. Using the desired transient time t_T and maximum overshooting $\bar{\sigma}$ given in Table 3 and the pole-placement approach [1, 2] design reference model (3.4), (3.5). If the maximum overshooting is zero, it worth selecting the binomial polynomial of the second order (since the plant of the second order and relative degree)

$$s^2 + 2\omega_0 s + \omega_0^2, \quad (3.16)$$

where ω_0 is the radius of the poles distribution calculated as

$$\omega_0 = \frac{t_T'}{t_T}$$

with normalized transient time t_T' of the system (here we assume $\omega_0' = 1$)

$$h(t) = \frac{1}{s^2 + 2s + 1} [1(t)].$$

If the maximum overshooting is zero, the Butterworth polynomial

$$s^2 + \sqrt{2}\omega_0 s + \omega_0^2$$

should be selected. In this case the normalized transient time for calculation of ω_0 is defined by the system

$$h(t) = \frac{1}{s^2 + \sqrt{2}s + 1} [1(t)] \quad (3.17)$$

To calculate the coefficients of the matrix A_M and the vector b_M ($n=2$) of the reference model we compare the companion polynomial of A_M with (3.16) or (3.17) depending on the choice.

Plot step response function of the reference model and show t_T with the overshooting $\bar{\sigma}$;

2. Based on assumption that the plant parameters are known design and simulate the closed-loop system with nonadaptive control (3.9) with the parameters defined according to (3.7) as

$$\theta_1 = \frac{-a_{m0} + a_0}{b_0}, \theta_2 = \frac{-a_{m1} + a_1}{b_0}.$$

Make three experiments with three sets of **plant** parameters:

- a_0 and a_1 ;
- negligibly change a_0 and a_1 parameters so that the system preserves the stability;
- change the parameters so that the system loses the stability.

Separately plot the elements of vector $x(t)$ together with the corresponding elements of $x_m(t)$. Plot the elements of $e(t)$.

3. Repeat this triple of experiments with adaptive controller (3.10) and the adaptation algorithm (3.13) instead of (3.9). Demonstrate properties 1-4 via simulation.

Make experiment with original parameters of the plant and three values of γ ;
Repeat the experiment for $g(t)=1$.

Separately plot the elements of vector $x(t)$ together with the corresponding elements of $x_m(t)$. Plot the elements of $e(t)$. Plot the elements of $\tilde{\theta}(t) = \theta - \hat{\theta}(t)$.

4. Make conclusions after each step of the work.

Table 3. The parameters of the plant and the reference model quantities

N_0	Matrix A	Coeff. b_0	Transient time, t_T	Maximum overshooting $\bar{\sigma}$, %	Reference $g(t)$
1	$\begin{bmatrix} 0 & 1 \\ -1 & -21 \end{bmatrix}$	7	0.4	<15	$\cos t + 2$
2	$\begin{bmatrix} 0 & 1 \\ -2 & -20 \end{bmatrix}$	8	0.6	<15	$2\text{sign}(\sin 0.4t) + 3$
3	$\begin{bmatrix} 0 & 1 \\ -3 & -19 \end{bmatrix}$	9	0.8	<15	$3\text{sign}(\sin 0.5t) + 3$

4	$\begin{bmatrix} 0 & 1 \\ -4 & -18 \end{bmatrix}$	1	0.6	<15	$2\sin 2t + \sin 5t + 8$
5	$\begin{bmatrix} 0 & 1 \\ -5 & -17 \end{bmatrix}$	2	0.4	<15	$4\text{sign}(\cos 2t) + 5$
6	$\begin{bmatrix} 0 & 1 \\ -6 & -16 \end{bmatrix}$	3	0.7	<15	$3\text{sign}(\cos 2t) + 3$
7	$\begin{bmatrix} 0 & 1 \\ -7 & -15 \end{bmatrix}$	4	0.5	<15	$\text{sign}(\sin t) + 2$
8	$\begin{bmatrix} 0 & 1 \\ -8 & -14 \end{bmatrix}$	5	0.6	<15	$0.8\sin 2t + \cos 0.8t + 2$
9	$\begin{bmatrix} 0 & 1 \\ -9 & -13 \end{bmatrix}$	6	0.2	<15	$\text{sign}(\sin 3t) + 1.5$
10	$\begin{bmatrix} 0 & 1 \\ -10 & -12 \end{bmatrix}$	7	0.4	<15	$7\text{sign}(\cos 0.9t) + 8$
11	$\begin{bmatrix} 0 & 1 \\ -11 & -11 \end{bmatrix}$	8	0.2	0	$\sin 3t + 1.5$
12	$\begin{bmatrix} 0 & 1 \\ -12 & -10 \end{bmatrix}$	9	0.2	0	$2\text{sign}(\cos t) + 3$
13	$\begin{bmatrix} 0 & 1 \\ -13 & -9 \end{bmatrix}$	1	0.4	0	$\cos t + 3\sin 2t + 5$
14	$\begin{bmatrix} 0 & 1 \\ -14 & -8 \end{bmatrix}$	2	0.5	0	$\text{sign}(\cos 2t)$
15	$\begin{bmatrix} 0 & 1 \\ -15 & -7 \end{bmatrix}$	3	0.7	0	$\text{sign}(\sin 0.5t) + 2$
16	$\begin{bmatrix} 0 & 1 \\ -16 & -6 \end{bmatrix}$	1	0.6	0	$10\cos 0.5t + 2\sin t + 12$
17	$\begin{bmatrix} 0 & 1 \\ -17 & -5 \end{bmatrix}$	2	0.7	0	$\text{sign}(\sin 0.5t) + 3$
18	$\begin{bmatrix} 0 & 1 \\ -18 & -4 \end{bmatrix}$	3	0.6	0	$0.5\text{sign}(\sin 0.7t) + 1$
19	$\begin{bmatrix} 0 & 1 \\ -19 & -3 \end{bmatrix}$	4	0.9	0	$5\sin 0.5t + 8$
20	$\begin{bmatrix} 0 & 1 \\ -20 & -2 \end{bmatrix}$	5	0.3	0	$\text{sign}(\cos t) + 3$
21	$\begin{bmatrix} 0 & 1 \\ -21 & -1 \end{bmatrix}$	6	0.2	<15	$\text{sign}(\sin 2t)$
22	$\begin{bmatrix} 0 & 1 \\ -22 & 0 \end{bmatrix}$	7	0.4	0	$6\text{sign}(\sin t) + 9$

23	$\begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix}$	8	0.6	0	$2\text{sign}(\sin t) + 4$
24	$\begin{bmatrix} 0 & 1 \\ 0.1 & -2 \end{bmatrix}$	9	0.2	0	$9\sin 2t + 9\cos t + 15$
25	$\begin{bmatrix} 0 & 1 \\ 0.2 & -3 \end{bmatrix}$	1	0.4	0	$4\text{sign}(\sin t) + 5$
26	$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$	2	0.3	0	$4\text{sign}(\cos t) + 3$
27	$\begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix}$	3	0.1	0	$\sin 0.1t + \cos 2t + 2$
28	$\begin{bmatrix} 0 & 1 \\ -2 & -9 \end{bmatrix}$	4	0.3	0	$9\text{sign}(\sin 2t) + 12$
29	$\begin{bmatrix} 0 & 1 \\ -3 & 0 \end{bmatrix}$	5	0.1	0	$3\text{sign}(\sin 3t) + 8$
30	$\begin{bmatrix} 0 & 1 \\ -9 & 0 \end{bmatrix}$	6	0.2	<15	$3\sin t + 3\cos t + 10$
31	$\begin{bmatrix} 0 & 1 \\ -8 & -2 \end{bmatrix}$	7	0.4	<15	$7\text{sign}(\cos t) + 8$
32	$\begin{bmatrix} 0 & 1 \\ 0.1 & 0 \end{bmatrix}$	8	0.2	0	$\sin 9t + 1$
33	$\begin{bmatrix} 0 & 1 \\ 0.2 & -1 \end{bmatrix}$	9	0.2	0	$2\text{sign}(\sin t \cos 2t) + 3$
34	$\begin{bmatrix} 0 & 1 \\ -10 & -1 \end{bmatrix}$	1	0.4	0	$\cos t - \sin 2t + 2$
35	$\begin{bmatrix} 0 & 1 \\ -9 & -1 \end{bmatrix}$	2	0.5	0	$\text{sign}(\cos 2t) + \text{sign}(\sin t)$
36	$\begin{bmatrix} 0 & 1 \\ -7 & -7 \end{bmatrix}$	3	0.7	0	$2\text{sign}(\sin t) + 4$
37	$\begin{bmatrix} 0 & 1 \\ -10 & 0 \end{bmatrix}$	1	0.6	0	$8\cos 2t + 3\sin t + 3$
38	$\begin{bmatrix} 0 & 1 \\ -9 & -2 \end{bmatrix}$	2	0.7	0	$2\text{sign}(\sin t) - 2$
39	$\begin{bmatrix} 0 & 1 \\ 0 & -19 \end{bmatrix}$	3	0.6	0	$-\text{sign}(\sin t) + 1$
40	$\begin{bmatrix} 0 & 1 \\ -27 & 0 \end{bmatrix}$	4	0.9	0	$5\sin t \cos 3t + 5$
41	$\begin{bmatrix} 0 & 1 \\ -9 & 0.2 \end{bmatrix}$	6	0.8	<15	$3\sin t + 3\cos t + 10$

42	$\begin{bmatrix} 0 & 1 \\ -8 & -2 \end{bmatrix}$	7	0.7	<15	$7\text{sign}(\cos t) + 8$
43	$\begin{bmatrix} 0 & 1 \\ -2 & 0 \end{bmatrix}$	8	0.6	0	$\sin 9t + 1$
44	$\begin{bmatrix} 0 & 1 \\ -6 & 0.3 \end{bmatrix}$	9	0.5	0	$2\text{sign}(\sin t \cos 2t) + 3$
45	$\begin{bmatrix} 0 & 1 \\ -10 & -1 \end{bmatrix}$	1	6	0	$\cos t - \sin 2t + 2$
46	$\begin{bmatrix} 0 & 1 \\ -9 & -0.5 \end{bmatrix}$	2	5	0	$7\text{sign}(\cos 2t) + 8$
47	$\begin{bmatrix} 0 & 1 \\ -7 & 0.1 \end{bmatrix}$	3	4	<15	$2\sin t + 4$
48	$\begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix}$	1	3	<15	$8\cos 2t + 3\sin t + 13$
49	$\begin{bmatrix} 0 & 1 \\ -3 & -3 \end{bmatrix}$	2	2	<15	$-2\sin t + 4$
50	$\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$	3	1	<15	$7\text{sign}(\sin t) + 7$

Typical questions

1. Does the condition (3.14) hold for the case of scalar plant in Work №1 if $g(t)=1$?
2. Can the solution of Lyapunov equation (3.12) be negatively definite matrix P ? Please clarify.
3. Can the solution of Lyapunov equation (3.12) be a diagonal matrix P ? Please clarify.
4. For which γ the adaptive system is unstable?
5. Give an example of 2-dimensional vector $x(t)$ not satisfying the persistent excitation condition (3.14).

Problems for 12 pts

1. Solve the problem of stabilization ($g(t) \equiv 0$) for the plant

$$\begin{aligned}\dot{x}_1 &= x_2, \\ \dot{x}_2 &= \theta_1 x_1 + \theta_2 x_1^3 + u,\end{aligned}$$

where $\theta_1 = 2$, $\theta_2 = 5$ are unknown parameters. For the reference model use zero overshooting and transient time $t_T = 1\text{sec}$. Simulate the adaptive system

2. Solve the control problem for the plant

$$\begin{aligned}\dot{x}_1 &= x_2, \\ \dot{x}_2 &= \theta_1 x_1 x_2 + \theta_2 (x_1^2 + 1) + u,\end{aligned}$$

where $\theta_1 = 2$, $\theta_2 = 5$ are unknown parameters. The objective is defined by equality (3.3). For the reference model use zero overshooting and transient time $t_T = 1\text{sec}$. The reference signal is given as $g(t) = 2\text{sign}(\sin t) + 1$. Simulate the adaptive system

3. Solve the control problem for the plant

$$\begin{aligned}\dot{x}_1 &= x_2, \\ \dot{x}_2 &= \theta x_2^2 + u,\end{aligned}$$

where θ is unknown parameter. The objective is defined by equality (3.3). For the reference model use zero overshooting and transient time $t_T = 1\text{sec}$. The reference signal is given as $g(t) = \sin t + 2$. Simulate the adaptive system

4. Solve the control problem for the plant

$$\begin{aligned}\dot{x}_1 &= 3x_2, \\ \dot{x}_2 &= \theta_1 x_1 + \theta_2 x_2 + u,\end{aligned}$$

where $\theta_1 = 1$, $\theta_2 = 2$ are unknown parameters. The objective is defined by equality (3.3). For the reference model use zero overshooting and transient time $t_T = 1\text{sec}$. The reference signal is given as $g(t) = 3\text{sign}(\sin 2t) + 2$. Simulate the adaptive system

5. Solve the control problem for the plant

$$\begin{aligned}\dot{x}_1 &= x_2, \\ \dot{x}_2 &= \theta_1 x_1 + \theta_2 x_2 + \theta_3 u,\end{aligned}$$

where $\theta_1, \theta_2, \theta_3 \geq \theta_{03} > 0$ are unknown coefficients. The objective is defined by equality (3.3). For the reference model use zero overshooting and transient time $t_T = 1\text{sec}$. The reference signal is given as $g(t) = \text{sign}(\sin t) + 2$. Simulate the adaptive system